

Reading the Validation Scatter Plot

diabetes-cerna-rnaseq · a learner's guide (Track A vs Track B, DP003)

Part 1 — What this specific plot shows

The figure is `results/figures/validation_scatter.png`: a comparison of **our pipeline's gene counts (Track A)** against the **published gene counts (Track B)** for one donor, **DP003**.
Headline result: **Pearson $r = 0.975$** .

Why it says “DP003”

The plot shows **one representative donor**, not all 12. In the script we picked DP003 (which is sample SRR13380421, a T2D donor) as the example.

Why just one? If all 12 donors were plotted on top of each other, the picture would be a smear. Showing one donor's genes gives a clean stand-in — and because all 12 donors had almost identical correlations (~ 0.974), this one fairly represents the whole batch. Any donor could have been chosen; DP003 was simply first.

Every component, labeled

- **Title** — *what* is compared (Track A vs Track B) and *for whom* (DP003), plus the headline number (Pearson $r = 0.975$).
- **X-axis (bottom)** — the **published** count for each gene, as `log2(count+1)`.
- **Y-axis (left)** — **our** count for the same gene, same scale.
- **Each hexagon** — a bucket of genes. With $\sim 40,000$ genes, individual dots would overlap, so nearby genes are grouped into hexagons. One hexagon = “genes that had roughly this published count and this our-count.”
- **Color + colorbar (right)** — *how many* genes are in each hexagon. Yellow/green = thousands stacked there; purple = only a few. The scale is logarithmic.
- **Red dashed line ($y = x$)** — the “perfect agreement” reference. A gene sitting exactly on it means our count equals their count.
- **The cloud of hexagons** — the data: every gene placed by its two counts.

Why log2 axes?

Gene counts range from 0 to millions. On a normal scale, a few huge genes would squash everything else into the corner. Taking log2 compresses that range so you can see *all* genes at once — the quiet ones and the loud ones.

Reading this plot's result

- The **dense green/yellow cloud rides along the red $y = x$ line** — the bulk of genes agree between pipelines. This is the visual form of $r = 0.975$.
- The **bright blob at (0,0)** = the many genes that are off/barely expressed in both. They agree these are silent. Normal.
- The **high end (top-right) sits on the line** = the most-expressed genes match almost exactly. The most important region, and it's tight.
- The **two faint edge streaks** (vertical at $x = 0$, horizontal at $y = 0$) = genes that are zero in one pipeline but not the other. These are **annotation differences**: we used Ensembl release 110, the authors used release 92, so a few genes are defined/counted differently. Small, explainable, and the reason r is 0.975 rather than a suspicious 1.0.

Verdict: cloud hugs the diagonal, streaks are thin, $r = 0.975$ -> a textbook “my pipeline reproduces the published result” figure.

Part 2 — How to read ANY scatter plot like a pro

A scatter plot shows the **relationship between two numbers**. Each point is one thing measured twice (here: one gene, measured by two pipelines). Read it in this order every time:

1. **Read the axes first.** What's on x, what's on y, and what's the scale (normal or log)? Never interpret before you know this. (*Here: their count vs our count, log scale.*)
2. **Look at the overall direction.** Do points rise together (x up, y up = **positive**), fall (**negative**), or scatter randomly (**no relationship**)? (*Here: they rise together — strong positive.*)
3. **Judge the tightness.** Points hugging a straight line = **strong**; a fat, spread-out cloud = **weak**. Tightness is what the correlation number r measures: **+1 = perfect rise, 0 = none, -1 = perfect fall**. (*Here: 0.975 = very tight.*)
4. **Use any reference line.** If there's a $y = x$ line, points *on* it mean the two measurements match. Above it = y higher; below it = x higher. (*Here: cloud sits on the line = pipelines agree.*)
5. **Spot the outliers.** Points far from the main pattern. Ask *why* — real signal or an artifact? (*Here: the faint edge streaks = annotation-version differences, explainable.*)
6. **Notice density (if shown).** Where do *most* points live? Color or darkness tells you. (*Here: the bright diagonal band = the vast majority of genes agree.*)

The one-sentence pro habit

Axes -> direction -> tightness -> reference line -> outliers.

Run through those five every time and you'll read any scatter plot correctly — gene counts, height vs weight, price vs sales, anything.

For *this* plot the method gives: two counts, rising together, very tightly, sitting right on the agreement line, with only tiny explainable outliers -> **your pipeline reproduces the published data.**